

Final project

HD 1: Optimal voluntary liquidity buffer

Bank k , currency c . The bank chooses μ_k^c before the liquidity shock and takes interbank/public coverage as given

1. Liquidity position

$$\chi_k^c = \mu_k^c + \chi p^c, \text{ where } \chi \begin{cases} 0, & \text{NU} \\ 1, & \text{U} \end{cases}$$

The liquidity shock

$$w_k^c = \beta_k A^c + e_k^c, \quad e_k^c \sim U[-a_k^c, a_k^c]$$

So, for one unit of deposit

$$\text{liquidity position} = \beta_k A^c + e_k^c + \mu_k^c + \chi p^c$$

To simplify:

$$z = \beta_k A + \mu + \chi p$$

2. Expected deficit

Deficits occurs when

$$z + e < 0$$

Then,

$$L(\mu, A) = E_e [(-z - e)^+ | A]$$

In the inner region of the support

$$-a < z < a,$$

deficits occurs for

$$e < -z$$

As $e \sim U[-a, a]$, its density is $1/2a$. So:

$$L(\mu, A) = \frac{1}{2a} \int_{-a}^{-z} (-z - e) de$$

$$\begin{aligned}
&= \frac{1}{2a} \left[-ze - \frac{e^2}{2} \right] - a \\
&= \frac{1}{2a} \left[z^2 - \frac{z^2}{2} - za + \frac{a^2}{2} \right] \\
&= \frac{1}{2a} \left[\frac{z^2}{2} - za + \frac{a^2}{2} \right] \\
&= \frac{1}{2a} \left[\frac{z}{\sqrt{2}} + \frac{a}{\sqrt{2}} \right]^2 \\
&= \frac{1}{2a} \left(\frac{1}{\sqrt{2}} \right)^2 (z+a)^2 \\
&= \frac{1}{4a} (z+a)^2
\end{aligned}$$

3. Deficit probability

In the same region:

$$\pi^-(\mu, A) = P(e < -z)$$

As e distributes uniformly

$$\pi^-(\mu, A) = \frac{e(z) - e(a)}{2a} = \frac{a-z}{2a}$$

Now, we take $\partial L / \partial \mu$

$$\frac{\partial L}{\partial \mu} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial \mu} = \frac{1}{2a} (z+a) \times 1 = \frac{1}{2a} (z+a) = - \frac{a-z}{2a}$$

But

$$\frac{a-z}{2a} = \pi^-(\mu, A)$$

So

$$\frac{\partial L(\mu, A)}{\partial \mu} = -\pi^-(\mu, A)$$

One extra unit of voluntary liquidity reduces the expected deficit exactly in the states in which the bank would otherwise be short.

4. Bank's problem

$$\min_{0 \leq \mu \leq \bar{\mu}} V(\mu) = K\mu + E_A [h(A) L(\mu, A)]$$

where:

K = opportunity cost of voluntary liquidity

$h(A)$ = private cost per unit of liquidity deficit

Key assumption: $h(A)$ is given for each bank

FOC:

$$\frac{\partial V}{\partial \mu} = K + E_A \left[h(A) \frac{\partial L(\mu, A)}{\partial \mu} \right]$$

$$\frac{\partial V}{\partial \mu} = K - E_A [h(A) \bar{\pi}(\mu, A)]$$

For an inner optimum:

$$V(\mu^*) = 0$$

So

$$K = E_A [h(A) \bar{\pi}(\mu^*, A)]$$

Marginal opportunity cost of self-insurance = Expected marginal deficit cost avoided
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HD2: Public dollar banking and self-insurance

1. Deficit cost

$$h(A) = \psi(A)q_I + [1 - \psi(A)][\phi(A)q_F + (1 - \phi(A))l]$$

Simplifying

$$h(A) = \psi q_I + [1 - \psi][l - \phi(l - q_F)]$$

Where:

ψ = interbank coverage

ϕ = public coverage of residual deficit

q_I = interbank funding cost

q_F = public funding cost

l = cost of uncovered deficit

2. Effect of R_D on coverage

For dollars

$$\phi(A) = \min \left\{ 1, \frac{F^M}{\Delta^M(A)} \right\}$$

In a state where the fund is rationed:

$$0 < F^M < \Delta^M(A)$$

So

$$\phi(A) = \frac{F^M}{\Delta^M(A)}$$

And

$$\frac{\partial F^M}{\partial R_D} = 1$$

Keeping $\Delta^M(A)$ fixed

$$\frac{\partial \phi(A)}{\partial R_D} = \frac{1}{\Delta^M(A)} > 0$$

(conditional comparative static:
 P, Q, Δ are held fixed)

in a rationed state

3. Effect on deficit cost

$$h(A) = \psi q_I + [1 - \psi][l - \phi(l - q_F)]$$

$$\frac{\partial h(A)}{\partial R_D} = -[1 - \psi][l - q_F]\phi_{RD}$$

Conditions:

$$\begin{aligned}\psi &< 1 \\ l &> q_F \\ \phi_{RD} &> 0\end{aligned}$$

So $\frac{\partial h(A)}{\partial R_D} < 0$ in any relevant and retained state

More public banking makes a residual liquidity shortage less costly to the bank

4. Implicitly differentiate the FOC

As

$$K = E_A [h(A, R_D) \pi^-(u^*, A)]$$

Define

$$G = K - E_A [h(A, R_D) \pi^-(u^*, A)] = 0$$

$$G_{RD} = -E_A [h_{RD}(A) \pi^-(u^*, A)]$$

Since

$$\frac{\partial h(A)}{\partial R_D} < 0 \text{ and } \pi^- \geq 0$$

So:

$$G_{RD} > 0$$

if exist at least one relevant state where the effect is strict

We know:

$$\frac{\partial \pi^-}{\partial u} = -f(-z(A))$$

So

$$G_u = -E_A \left[h(A) \frac{\partial \pi^-}{\partial u} \right]$$

$$G_{\mu} = E_A[h(A) f(-2|A)]$$

Define

$$D \equiv E_A[h(A) f(-2|A)] > 0$$

By the implicit function theorem:

$$\frac{d\mu^*}{dR_D} = - \frac{G_{R_D}}{G_{\mu}}$$

Substituting

$$\frac{d\mu^*}{dR_D} = \frac{E_A[h R_D(A) \pi^-(\mu^*, A)]}{D}$$

Replacing h as:

$$\frac{d\mu^*}{dR_D} = \frac{-(1-q_F) E_A[\pi^-(\mu^*, A) (1-\psi(A)) \phi_{RD}(A)]}{D}$$

Since:

$$1 > q_F, D > 0$$

and at least one state where:

$$\pi^- > 0, \quad \psi < 1, \quad 0 < P^{\mu} < \Delta^{\mu},$$

finally we obtain

$$\frac{d\mu^*}{dR_D} < 0$$

More public dollar backing lowers the private cost of being short. Therefore the bank has less incentive to self-insure with a voluntary buffer

Scope: This is a conditional individual best-response result. It is not yet a general-equilibrium result because P, α, Δ, ψ may change when all bank adjust their buffers